

Labwork 8: Scatter

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1 RGB to HSV

I use the RGB image from Lab 3. Each 2D thread divides its three channels by 255 and applies the formulas in the slide. Hue is in degrees; saturation and value are between 0 and 1. I store H, S and V in three separate arrays instead of keeping three channels together for each pixel. For gray pixels I set hue to 0; for black pixels I also set saturation to 0.

2 HSV to RGB

I select one of the six hue sectors and calculate the three RGB channels. I multiply by 255, round to the nearest integer and keep values in 0 to 255. Both kernels use 16 by 16 blocks and check image boundaries.

3 Comparison

Maximum channel difference: 0
Different pixels: 0

The restored image matches the input exactly in this test. I compare decoded pixels and save the output as PNG to avoid JPEG changes.



Input RGB image and RGB image restored from HSV.